

Task Completion Human Rubric

Focus strictly on task completion. Ignore visual glitches, clipping, or impossible physics, provided the macro-attempt and intent are clearly recognizable. Only fail a gate if it's a clear NO, or a considerable physical error makes it impossible / contradictory to confirm that the intent/task/state was actually attempted or accomplished.

Input → Video + Goal | Output → Float[0-3]

[Divide High-Level Goal into N Subgoals with Dependencies]
Output: S1 (no prereq), S2 (requires: S1), S3 (requires: S1)...

For Each Subgoal i (1 to N)

Q1: Is there clear initiation/behavior in scene that attempt to solve this specific subgoal?

└ NO → [Score_i = 0]
└ YES

Q2: Is the intent followed by a high-level execution end-to-end approaching the goal?

└ NO → [Score_i = 1]
└ YES

Q3: Does the process end with the intended final state?

└ NO → [Score_i = 2]
└ YES → [Score_i = 3]

[Calculate Base_Score = MEAN(Score_1 ... Score_N)]

Is Base_Score >= 2.0?

└ NO → Final Task Score = Base_Score
└ YES

Q4: Were prerequisite subgoals completed before their dependent actions began?

└ YES → Final Task Score = Base_Score
└ NO → Final Task Score = 1.0

Subgoal Decomposition

- a. Given a high level goal, split into [1,N] subgoals
- b. Subgoal should include only 1 source + at most 1 target + 1 action
 - i. Bottle is moved inside the bag → **TRUE**
 - ii. Fork and spoon is moved on top of the tray → **FALSE**
 1. Should be splitted further to
 - a. Fork is moved on top of the tray
 - b. Spoon is moved on top of the tray
 - iii. Bag is zipped → **TRUE**
- c. Example

Scene	High-level Goal	Subgoals	Deps
	A dumbbell and water bottle stored inside the brown paper bag, with the filled paper bag placed inside the laundry basket, and the rugby ball positioned on the empty spot previously occupied by the paper bag.	- dumbbell is moved into the brown paper bag - water bottle moved into the brown paper bag - paper bag moved into laundry basket - rugby ball is placed on top of the green cabinet	$S1 \leftarrow \emptyset$ $S2 \leftarrow \emptyset$ $S3 \leftarrow S1, S2$ $S4 \leftarrow S3$
	Pour the water from a closed bottle half to the metal pot and half to the yellow mug, then close the mug with the metal cap	- Open water bottle - Pour half of the water to metal pot - Pour half of the water to yellow mug - Close the yellow mug with metal cap	$S1 \leftarrow \emptyset$ $S2 \leftarrow S1$ $S3 \leftarrow S1$ $S4 \leftarrow S2, S3$

Q1: Intent

- a. Is there any task-directed behavior involving the correct source object, tool, or mechanism that was initiated?
- b. Core heuristic
 - i. **Pass (YES):** Hand/agent travels to item, engages pre-grasp, and is sensible to become the initiation of task completion attempt/process.
 - ii. **Fail (NO):** Hand moves away, stays still, or targets the wrong item, or **physics error makes it impossible to verify** → **FALSE (Score = 0)**

ONLY USE BELOW SECTION IF PHYSICS ERROR / GLITCH YOU ARE ENCOUNTERING MAY AFFECT YOUR JUDGEMENT

A physics or rendering error should **ONLY** fail anything here if the error **completely destroys or contradicts**. If the evidence is visually verifiable despite the glitch, **PASS the Task Gate** and log the error exclusively in the Physics Rubric.

- c. **It supposed to be independent of physics error**, but if there is one, try your best to use these heuristics:
 - i. Morphing hands, clipping, 6 fingers, or floating items / imprecise “contact” → **IGNORE (Q1 remains TRUE)**
 - ii. Object duplication / spawned instances / flickery → **REMAINS TRUE (Log glitch in Physics Rubric)**

Visual Scene	Subgoal	Video Gen Behavior that might imply physics error	Q1 Result
Morphing Hand	Open water bottle	Hand morphs into a blurry 6-fingered shape, but extends directly toward the bottle cap.	TRUE (Go to Q2)
Clipping Reach	Dumbbell is moved into bag	Hand clips straight through the table surface while reaching out to grasp the dumbbell.	TRUE (Go to Q2)
Object Duplication	Pour water into metal pot	Water bottle duplicates on screen; hand reaches toward the spawned duplicate bottle.	TRUE (Go to Q2)

Disappearing Object	Hand bottle	grabs	If the bottle vanishes although the agent/hand is already going to the bottle (not because of occlusion by hand!) its counted as TRUE here, because intent is achieved	TRUE (Go to Q2)
Wrong Target	Open bottle	water	Hand extends from the bottom periphery and reaches directly for the yellow mug instead.	FALSE (Score = 0)

Q2: High-level Process

- a. Is the intent followed by a high-level action trajectory that is executed across space/time end-to-end?
- b. **Intent vs. Process boundary**
 - i. Intent (Q1): Reaching toward, touching, or orienting to the target object (imagine like a setup before the actual action/process)
 - ii. Process (Q2): Carrying out the high-level trajectory attempt from start to finish, the finish itself is no need to be precise, just high-level (the actual action)
- c. Core heuristics
 - i. **Pass (YES):** The characteristic trajectory (carrying, pouring, wiping, turning, etc.) is substantially executed till the to be verified end state later in Q3.
 - ii. **Fail (NO):** Motion stops after setup, item is dropped immediately, or trajectory goes in wrong direction, simply fails, or **physics error makes it impossible to verify**

ONLY USE BELOW SECTION IF PHYSICS ERROR / GLITCH YOU ARE ENCOUNTERING MAY AFFECT YOUR JUDGEMENT

A physics or rendering error should **ONLY** fail anything here if the error **completely destroys or contradicts**. If the evidence is visually verifiable despite the glitch, **PASS the Gate** and log the error exclusively in the Physics Rubric.

- d. **It is supposed to be independent of physics error**, but if there is one, try your best to use these heuristics:
 - i. Floating items mid-transit, clipping through obstacles, flickery,, or unnatural speed → **IGNORE** (Q2 remains TRUE)
 - ii. Object teleports/snaps directly to target area during movement → REMAINS **TRUE** (Log glitch in Physics Rubric)
 - iii. Action completely aborted mid-flight or trajectory goes in opposite direction / **complete collapse** → **FALSE** (Score = 1)

Visual Scene	Subgoal	Video Gen Behavior that might imply physics error	Q2 Result
Floating Transit	Water bottle is moved into bag	The bottle floats 2 cm away from the hand as it travels across space into the bag.	TRUE (Go to Q3)

Mesh Clipping	Dumbbell moved into bag	is	Dumbbell clips through the rim of the bag mid-transit before settling inside.	TRUE (Go to Q3)
Teleport Snap	Yellow mug closed with cap	is	Cap snaps/teleports onto the mug rim halfway through the hand's reach movement.	TRUE (Go to Q3)
Aborted Motion	Water bottle opened	is	Hand touches the cap and begins twisting, but stops after a tiny turn and retracts.	FALSE (Score = 1)
Dropped Misdirected	Water bottle moved into bag	is	Hand lifts bottle, but moves it sideways toward the floor and releases it far from the bag.	FALSE (Score = 1)

Q3: Final State Verification

- a. Does the process end with the target object resting in the exact required final state specified by the subgoal?
- b. Motion Trajectory vs. Final State adherence boundary:
 - i. **Q1 & Q2 (Intent & Process):** Leniency applied to visual glitches, clipping, and rendering noise during movement.
 - ii. **Q3 (Final State): Less leniency.** Inspecting the post-action evidence frame per subgoal, the physical outcome must confirm the goal state was reached.
- c. Core heuristic
 - i. **Pass (YES):** Target object satisfies the explicit prompt condition (position, relation, state).
 - ii. **Fail (NO):** Object dropped off-target, incomplete state, relation explicitly fails, or physics error makes it impossible to verify.

ONLY USE BELOW SECTION IF PHYSICS ERROR / GLITCH YOU ARE ENCOUNTERING MAY AFFECT YOUR JUDGEMENT

A physics or rendering error should **ONLY** fail anything here if the error **completely destroys or contradicts**. If the evidence is visually verifiable despite the glitch, **PASS the Gate** and log the error exclusively in the Physics Rubric.

- d. **It is supposed to be independent of physics error**, but if there is one, try your best to use these heuristics:
 - i. Final state is accurate, but contains minor background rendering noise (floating a bit, changing minor attribute), or flickery → **TRUE (Score = 3)**
 - ii. Object missing entirely or destroyed at the final frame → **FALSE (Score = 2)**

Visual Category	Scene Subgoal	Final Frame Visual State	Q3 Result
Fluid Transfer	Water is poured into metal pot	Liquid level inside the metal pot visibly increased / filled to target.	TRUE (Score = 3)
State Transformation	Paper bag is folded	Paper bag is in a flattened, folded geometry resting on the table.	TRUE (Score = 3)

Tool / Surface Cleaning	Spill is wiped with sponge	Table surface area is clean / free of fluid stain where sponge passed.	TRUE (Score = 3)
Object Relocation	Dumbbell is placed on green mat	Dumbbell is lying completely outside the perimeter of the green mat.	FALSE (Score = 2)
Precision	Drawer is pulled open Fill half bottle	Drawer was tugged, but remains 90% closed (only cracked open 1 cm). Water is filled by instead of half, its being filled 70%, far from target preciseness (however 45-55% is okay → Score = 3)	FALSE (Score = 2)

Q4: Dependency check

- a. The heuristics is simple, for every dependency
 - i. $GOAL \leftarrow DEPS_1, DEPS_2, \dots, DEPS_N$
 - ii. If either GOAL or any of DEPS is not present in the video, skip checking Q4 for this dependency, go to next one / end scoring by choosing 'Yes'
 - iii. If all present, just check if the right hand side and left hand side is complete.
 1. If YES, then go to next deps / if nothing left, choose 'Yes'
 2. If NO, immediately flag Q4 as 'No'